{
"creator" : "ari.neko@cvut.cz",
"timestamp" : "09-07-2026_17:45:35",
"variant" : 2,
"subjectName" : "computer-networks",
"categories" : ["Routing Protocols", "Transport Layer (TCP/UDP)", "IP Addressing & Subnetting"],
"questions" : [{
"question" : "What classification defines the Open Shortest Path First (OSPF) routing protocol engine?",
"answers" : ["Distance-Vector routing protocol", "Static lookup layout interface", "Path-Vector configuration format", "Link-State routing protocol"],
"correctIndex" : 3
}, {
"question" : "Which of the following routing protocols operates as an Exterior Gateway Protocol (EGP) to handle internet traffic?",
"answers" : ["Enhanced Interior Gateway (EIGRP)", "Routing Information Protocol (RIP)", "Open Shortest Path First (OSPF)", "Border Gateway Protocol (BGP)"],
"correctIndex" : 3
}, {
"question" : "What algorithm forms the mathematical engine used by OSPF to compute shortest path trees?",
"answers" : ["Banker's Algorithm", "Floyd-Warshall Routine", "Bellman-Ford Algorithm", "Dijkstra's Algorithm"],
"correctIndex" : 3
}, {
"question" : "What metric parameter does the Routing Information Protocol (RIP) utilize to determine the best path?",
"answers" : ["Hop count", "Bandwidth capacity", "Total circuit delay latency", "Link cost speed calculations"],
"correctIndex" : 0
}, {
"question" : "What occurs when a TCP host triggers a Reset (RST) flag parameter inside its header field?",
"answers" : ["The sender requests a temporary pause in traffic flow", "The connection is instantly terminated due to an unrecoverable error mismatch condition", "The receiver reports that a data segment contains corrupt checksum blocks", "The pipeline starts a context synchronization sequence"],
"correctIndex" : 1
}, {
"question" : "Which of the following traits characterizes the User Datagram Protocol (UDP) data stream?",
"answers" : ["Operates exclusively inside Layer 3 routing tracks", "Reliable, connection-oriented,

and features heavy window sizing flows", "Mandates manual packet handshakes before transmission", "Connectionless, lightweight, and lacks delivery guarantees or sequencing"],
"correctIndex" : 3
}, {
"question" : "What does the 'Sliding Window' mechanism accomplish inside a TCP session tracking pipeline?",
"answers" : ["Cryptographic block rotations to shield payload fields", "Horizontal expansion of port bounds when traffic spikes occur", "Automatic file segment translations", "Dynamic flow control to regulate how much data a sender can transmit before an ACK is required"],
"correctIndex" : 3
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"question" : "What flag sequence handles a standard connection teardown/closure inside TCP transport logic?",
"answers" : ["RST, ACK", "FIN, ACK, FIN, ACK", "SYN, SYN-ACK, ACK", "PSH, ACK, URG"],
"correctIndex" : 1
}, {
"question" : "If a host configuration has an IP of 192.168.1.50 with a subnet mask of /26, how many hosts are available?",
"answers" : ["64 hosts", "30 hosts", "62 hosts", "126 hosts"],
"correctIndex" : 2
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"question" : "What is the primary role of the Address Resolution Protocol (ARP) inside a subnet?",
"answers" : ["To assign dynamic IP configurations automatically", "To map text-based domain strings", "To discover the physical MAC address matching a known logical IP address", "To forward data packages across different global public networks"],
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